ECO 202 — Practice Exam 1 — Version B

Statistics and Data Analysis for Economics | Princeton University

Fall 2026 | Classes 1–5 | 100 points

Name: ______________________ Student ID: ______________

Timing. Target workload: 70 minutes (10 minutes multiple choice; 60 minutes written work). Follow announced start/end times within the 80-minute meeting. For practice, set a 70-minute timer.

Conditions. No books, notes, calculators, AI, or other electronics; University-authorized accommodations apply. Fractions, radicals, and equivalent exact expressions are acceptable. Show reasoning in written answers. Data are teaching examples unless identified otherwise.

Points. Questions 1–4: one answer A–D, 5 points each, no negative marking. Questions 5–8: 20 points each; subpart points are shown. Label any continuation clearly.

Multiple choice (20 points; about 10 minutes)

1. A dataset records the three-digit telephone area code of each customer. Which description fits area code?

A. Quantitative, because it is written using digits.

B. A measure of the amount of telephone service.

C. A continuous outcome.

D. Categorical, because the numbers label groups.

2. Under a continuous density model, the fraction of values between a and b is represented by:

A. the sum of the density heights at a and b.

B. the area under the density between a and b.

C. the density height at the midpoint.

D. the interval length, regardless of the density.

3. A regression predicts weekly sales in thousands of dollars from store area in hundreds of square feet. Which describes the slope's units?

A. Thousands of dollars per hundred square feet.

B. Hundreds of square feet per thousand dollars.

C. No units; every regression coefficient is standardized.

D. Thousands of dollars squared.

4. For a treated worker with an observed outcome $Y_i(1)$, the missing counterfactual is:

A. the same worker's outcome under no treatment, $Y_i(0)$.

B. the observed mean of the treated workers.

C. the outcome of any untreated worker.

D. the worker's outcome measured after treatment.

5. A boxplot and a change of scale (20 points; about 15 minutes)

Six values are $1, 3, 3, 5, 7, 8$. Use the median-of-halves quartile convention. For variance you may use $\sum x_i^2 = 157$ and $s^2 = [\sum x_i^2 - n\bar{x}^2]/(n-1)$.

(a) (6 points) Calculate the median, quartiles, and IQR. Sketch a labeled boxplot with whiskers extending to the extreme observations within the 1.5IQR fences.

(b) (6 points) Calculate the mean and sample variance. State how the units of variance differ from the units of the observations.

(c) (4 points) Transform each value by $w = 20 - 2x$. Find the transformed median and IQR without listing every new value. Explain how the negative multiplier changes the ordering and why the IQR remains nonnegative.

(d) (4 points) Another dataset has $Q_1 = 3$, $Q_3 = 7$, and an observation of 20. Apply the outlier rule and explain what you would need to check before deleting that observation.

6. Transformations and model-based comparisons (20 points; about 15 minutes)

(a) (6 points) A density has constant height $1/4$ on $[-2, 2]$ and is zero elsewhere. Sketch it and find its median. What fraction of its values satisfies $(X+2)/4 < 3/4$? Show the transformed cutoff and area.

(b) (5 points) In a separate teaching model, delivery time T follows $\mathsf{N}(30, 5^2)$ in minutes. Find the fraction below 20 minutes using $\Phi(-2) = 0.0228$. Explain why a model standard deviation of 25 would be incorrect here.

(c) (5 points) Company A's delivery times are Normal with mean 30 and standard deviation 5; Company B's are Normal with mean 40 and standard deviation 10. Compare a 35-minute delivery at A with a 45-minute delivery at B, first in minutes and then in percentile position. Use $\Phi(1) = 0.8413$ and $\Phi(0.5) = 0.6915$.

(d) (4 points) An analyst claims that 95% of every dataset must lie within roughly two standard deviations of its mean. Explain the problem and identify one graphical feature that would lead you to question a Normal model.

7. A small scatterplot and its residuals (20 points; about 15 minutes)

Three observed pairs are $(x, y) = (0, 1), (1, 3), (2, 2)$, with x measured in hours and y in units of output. You may use

$$S_{xx} = \sum(x_i - \bar{x})^2, \quad S_{yy} = \sum(y_i - \bar{y})^2, \quad S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y}),$$

$$b_1 = S_{xy}/S_{xx}, \qquad b_0 = \bar{y} - b_1\bar{x}, \qquad r = S_{xy}/\sqrt{S_{xx}S_{yy}}.$$

(a) (6 points) Find the two means, the three centered sums, and the least-squares line with an intercept.

(b) (4 points) Sketch the three observations and fitted line on labeled axes. Mark the vertical residual for the point $(1, 3)$ and calculate it.

(c) (6 points) Calculate r, r^2, and all three residuals. Check that residuals sum to zero. Explain what r^2 measures.

(d) (4 points) A fourth observation with $x = 50$ is added. Explain why it has high leverage and how you would assess whether it is influential. Can its effect on the fitted slope be determined from its x value alone?

8. Group composition and a causal claim (20 points; about 15 minutes)

The following fictional observational records describe training participants and nonparticipants. Prior skill was measured before training, and all outcomes are subsequent earnings in thousands of dollars.

	Training		No training	
Prior skill	Count	Mean earnings	Count	Mean earnings
High	1	12	3	10
Low	3	6	1	4

(a) **(6 points)** Calculate mean earnings among all participants and among all nonparticipants, and their difference. Show how the counts enter the calculation.

(b) **(4 points)** Calculate the participant-minus-nonparticipant mean difference separately within each prior-skill group. Compare the signs with part (a).

(c) **(6 points)** Explain how the group composition produces the contrast between (a) and (b). Does either comparison by itself establish a causal training effect? Explain what remains uncertain even after separating workers by prior skill.

(d) **(4 points)** For a worker who participated, define the missing counterfactual relevant to this question. Explain why another worker in the same prior-skill group need not supply that counterfactual exactly.